

Cat of CSC 1263 Discrete Structures for Computing

1st Year Computer and Software Engineering. Max: 30pts Duration: 1h. Answer all questions

Question 1. 5pts

The sum of the first 3 terms of an arithmetic series is 21 and the sum of the next three terms is 66.

- Find the value of the first term and the common difference.
- Write an expression for the n^{th} term of the series u_n .
- Find the sum of the first 10 terms.

Question 2. 4pts

Let $P(n)$ be the statement that $1^2 + 2^2 + \dots + n^2 = n(n+1)(2n+1)/6$ for the positive integer n .

- What is the statement $P(1)$?
- Show that $P(1)$ is true, completing the basis step of the proof.
- What is the inductive hypothesis?
- What do you need to prove in the inductive step?
- Complete the inductive step, identifying where you use the inductive hypothesis.

Question 3. 4pts

Find a formula for the n th term of the following geometric sequence. What is the ninth term of the sequence? 5, 15, 45, ...

Question 4. 4pts

A parking lot has 31 visitor spaces, numbered from 0 to 30. Visitors are assigned parking spaces using the hashing function, where k is the number formed from the first three digits on a visitor's license plate.

- Which spaces are assigned by the hashing function to cars that have these first three digits on their license plates: 317, 918, 007, 100, 111, 310?
- Describe a procedure visitors should follow to find a free parking space, when the space they are assigned is occupied.

Question 5. 5 pts

- Find 5 sequence number of pseudorandom numbers generated by the linear congruential method with modulus $m = 9$, multiplier $a = 7$, increment $c = 4$, and seed $x_0 = 3$.
- Find the solutions of the simultaneous system of recurrence relations $a_n = a_{n-1} + b_{n-1}$ $b_n = a_{n-1} - b_{n-1}$ with $a_0 = 1$ and $b_0 = 2$.

Question 6. 4pts

- Use the Euclidean algorithm to calculate $\gcd(100, 270)$. Show each step.
- Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology

Question 7. 4pts

- Encrypt the plaintext message "STOP GLOBAL WARMING" using the shift cipher with shift $k = 11$.
- Decrypt the ciphertext message "LEWLYPLUJL PZ H NYLHA ALHJOLY" that was encrypted with the shift cipher with shift $k = 7$

Good Luck!!!!